

Chathuranga Weeraddana

Senior Research Fellow (Docent), University of Oulu, Finland
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1. OBJECTIVES

My work spans **distributed optimization, distributed machine learning, signal processing, and wireless communications**, with a focus on the **theoretical and algorithmic foundations of learning and optimization for large-scale networked systems**. I aim to advance **rigorous methodological frameworks** while applying these approaches to **data-driven and interdisciplinary challenges**. Alongside my research, I am committed to **high-quality teaching, curriculum development, and student supervision**, integrating research-led insights into teaching to support student learning and engagement in AI and data-driven systems.

2. RESEARCH INTERESTS

Distributed optimization, distributed machine learning, multi-stage decision problems, surrogate modeling in optimization, signal processing, wireless communication.

3. TEACHING INTERESTS

Including, but not limited to: Mathematical optimization, machine learning, statistical signal processing.

4. DEGREES

University of Oulu	Finland
<i>Doctor of Science, Telecommunication Engineering</i>	2011
Thesis: Optimization Techniques for Radio Resource Management in Wireless Communication Networks	
Reviewers: Prof. Leandros Tassiulas (Yale University, USA), Prof. Lars K. Rasmussen (KTH, Sweden)	
Opponents: Prof. Lars K. Rasmussen (KTH, Sweden), Prof. Olav Tirkkonen (Aalto University, Finland)	

Asian Institute of Technology	Thailand
<i>Master of Engineering, Telecommunication Engineering</i>	2007
Thesis: Performance Analysis of OFDM Systems with Random Residual Frequency Offset	

University of Moratuwa	Sri Lanka
<i>Bachelor of Science, Electronic and Telecommunication Engineering</i>	2004

5. LEADERSHIP EXPERIENCES

- **DESIRE6G** (*Deep Programmability and Secure Distributed Intelligence for Real-Time End-to-End 6G Networks*), 2023-2025, funded under Horizon Europe / SNS JU
 - Managed the University of Oulu's contributions to the Horizon Europe DESIRE6G, coordinating technical activities, deliverables, and collaboration with consortium partners.
- **IEEE MERCon**, the flagship international conference of the University of Moratuwa, Sri Lanka
 - International Relations Chair, 2022
 - Publication Chair, 2020
- **NCTM, National Conference on Technology and Management**, organized by the Research Center, SLIIT, Sri Lanka
 - Co-Chair, 2016

6. RESEARCH AND PROFESSIONAL EXPERIENCE

CWC-RT, University of Oulu <i>Docent</i> Working in the subgroup Intelligent Connectivity and Networks/Systems Group (ICON)	Finland 2024/09 - To Date
CWC-RT, University of Oulu <i>Senior Research Fellow</i>	Finland 2022/08 - 2024/09
Dept. of Electronic & Telecommun. Engineering, University of Moratuwa <i>Senior Lecturer (equivalent to Associate Prof.)</i>	Sri Lanka 2018 - 2022/07
Dept. of Electrical & Computer Engineering, SLIIT <i>Senior Lecturer (equivalent to Associate Prof.)</i>	Sri Lanka 2015 - 2018
Dept. of Automatic Control, KTH Royal Institute of Technology <i>ACCESS Postdoc Researcher</i>	Sweden 2012-2014

7. SELECTED PUBLICATIONS

1. H. Abeynanda, **P. C. Weeraddana**, and C. Fischione, "On the Characteristics of the Conjugate Function Enabling Effective Dual Decomposition Methods," *IEEE Transactions on Signal Processing*, vol. 74, pp. 572-588, Jan. 2026.
2. T. AlShammari, **P. C. Weeraddana**, and M. Bennis, "BayGO: Decentralized Bayesian Learning and Information-Aware Graph Optimization Framework," *IEEE Transactions on Signal Processing*, vol. 72, pp. 2101-2116, Apr. 2024.
3. S. Ranaweera, **P. C. Weeraddana**, P. Dharmawansa, and C. Fischione, "On the Convergence of Inexact Gradient Descent With Controlled Synchronization Steps," *IEEE Signal Processing Letters*, vol. 30, pp. 703-707, May. 2023.
4. H. Abeynanda, **P. C. Weeraddana**, G. H. J. Lanel, and C. Fischione, "On the Primal Feasibility in Dual Decomposition Methods Under Additive and Bounded Errors," *IEEE Transactions on Signal Processing*, vol. 71, pp. 655-669, Feb. 2023.
5. M. Ding, S. Atapattu, **P. C. Weeraddana**, and J. Evans, "Energy-Efficiency Fairness of Interference Multi-Relay Networks for Multi-User Communication," *IEEE Communication Letters*, vol. 25, no. 3, Mar. 2021.
6. S. Magnússon, **P. C. Weeraddana**, M. Rabbat, and C. Fischione, "On the convergence of alternating direction lagrangian methods for nonconvex structured optimization problems," *IEEE Transactions on Control of Network Systems*, vol. 3, no. 3, pp. 296-309, Oct. 2016.
7. S. Magnússon, **P. C. Weeraddana**, and C. Fischione, "A distributed approach for the optimal power flow problem based on admm and sequential convex approximations," *IEEE Transactions on Control of Network Systems*, vol. 2, no. 3, pp. 238-253, Oct. 2015.
8. G. Athanasiou, **P. C. Weeraddana**, C. Fischione, and L. Tassiulas, "Optimizing client association for load balancing and fairness in millimeter-wave wireless networks," *IEEE/ACM Transactions on Networking*, vol. 23, no. 3, pp. 836-850, Mar. 2014.
9. **P. C. Weeraddana**, M. Codreanu, M. Latva-aho, A. Ephremides, and Carlo Fischione, "Weighted sum-rate maximization in wireless networks: A review," *Foundations and Trends in Networking, Now Publishers Inc.*, vol. 6, no. 1-2, pp. 1-163, 2012.
10. **P. C. Weeraddana**, M. Codreanu, M. Latva-aho, and A. Ephremides, "Weighted sum-rate maximization for a set of interfering links via branch and bound," *IEEE Transactions on Signal Processing*, vol. 59, no. 8, pp. 3977-3996, Aug. 2011.

11. **P. C. Weeraddana**, M. Codreanu, M. Latva-aho, and A. Ephremides, "Multicell downlink weighted sum-rate maximization: A distributed approach," *IEEE Transactions Signal Processing*, vol. 61, no. 3, pp. 556–570, February 2013.

8. Teaching Experience

Div. of Network and Systems Engineering, KTH Royal Institute of Technology <i>Lecturer, Graduate Course</i> MM Optimization Algorithms with Machine Learning Applications https://chathugithub1.github.io/Home/courses/courseinfo.html	Sweden 2022/03-2022/06
Faculty of Engineering, University of Moratuwa <i>Lecturer, Graduate Courses</i> Engineering Decision Theory, Broadband Communications, Digital Communications, Advanced Instrumentation	Sri Lanka 2018-2022
Faculty of Engineering, University of Moratuwa <i>Lecturer, Undergraduate Courses</i> Introduction to Telecommunication, Teletraffic Engineering	Sri Lanka 2018-2022
Dept. of Electrical & Electronics Engineering, University of Peradeniya <i>Lecturer, Graduate Course</i> Optimization in Communication Systems	Sri Lanka 2018- 2022
Dept. of Electronic Engineering, Yangon Technical University <i>Lecturer, Graduate Course</i> Introduction to Mathematical Optimization	Myanmar 2018 - 2019
Dept. of Electrical & Computer Engineering, SLIIT <i>Lecturer, Graduate Courses</i> Several Communication and Signal Processing Courses	Sri Lanka 2015 - 2018
Dept. of Electrical & Computer Engineering, Curtin University <i>Lecturer, Undergraduate Courses through SLIIT</i> Several Communication and Signal Processing Courses	Australia 2015 - 2018
Dept. of Automatic Control, KTH Royal Institute of Technology <i>Lecturer, Graduate Courses</i> Network Optimization, Principles of Wireless Sensor Networks	Sweden 2013-2014
Centre for Wireless Communications, University of Oulu <i>Lecturer, Graduate Courses</i> Convex Optimization with Applications in Wireless Communications and Networks	Finland 2010

9. Invited Talks and Visits for Research/Teaching

KTH Royal Institute of Technology <i>Guest Researchers</i>	Sweden 2022/03/01 - 2022/06/01
Dept. of Electrical & Electronics Engineering, University of Peradeniya <i>Visiting Lecturer</i>	Sri Lanka 2018-2022
Dept. of Electronic Engineering, Yangon Technical University <i>Visiting Lecturer</i>	Myanmar 2019/08/19 - 2019/09/01
Dept. of Electronic Engineering, Yangon Technical University <i>Visiting Lecturer</i>	Myanmar 2018/09/10 - 2018/09/20
16th International Conference on Industrial and Information Systems (IEEE ICIIS) <i>Guest Speaker</i> Session: 5G Evolution and 6G	Sri Lanka 2021

10. AWARDS AND HONOURS

- **ACCESS Post-Doctoral Fellowship:** Among 1 of the 3 post-doctoral fellows selected from ACCESS Linnaeus Centre, KTH Royal Institute of Technology, Sweden, 2012
- **Best doctoral thesis award:** Dept. of Comm. Eng., University of Oulu, Finland, 2011
- **Second best paper award:** Asilomar Conference on Signals, Systems and Circuits 2010
- **GETA grant:** Graduate School in Electronics, Telecommunications and Automation grant for a 4 years of graduate school position in University of Oulu, Finland, 2011
- **Students' research grants:** Tauno Tönning Säätiö, Finland, 2011, 2009
- **Students' research grants:** Oulun Yliopiston Tuksäätiö, Finland, 2010
- **Students' research grants:** Nokia Foundation, Finland, 2010, 2009
- **Students' research grants:** Tekniikan Edistämissäätiö Stiftelsen, Finland, 2008
- **The AIT alumni association prize:** in recognition of the most outstanding academic performance among students in the School of Engineering and Technology, AIT, Thailand, 2007
- **The Sonera prize:** in recognition of the most outstanding academic performance among students in Telecommunication, AIT, Thailand, 2007
- **King's Scholarship:** from His Majesty King Bhumibol Adulyadej of the kingdom of Thailand, 2005

11. OTHER KEY ACADEMIC MERITS

- **Exemplary Reviewer:** Certificate of appreciation from IEEE Communication Society, IEEE Wireless Communications Letters, 2020,2021
- **Reviewer:** IEEE Transactions of Communications, IEEE Wireless Communications Letter, IEEE Transactions of Signal Processing, and IEEE Transactions of Wireless Communications, among others
- Completed an Induction Training course at the Staff Development Centre, Open University of Sri Lanka (2019), covering key pedagogical frameworks including outcomes-based education (OBE), constructive alignment, scaffolding, constructivism, curriculum design and mapping, program design, and assessment methods.

Referees

Professor Carlo Fischione

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